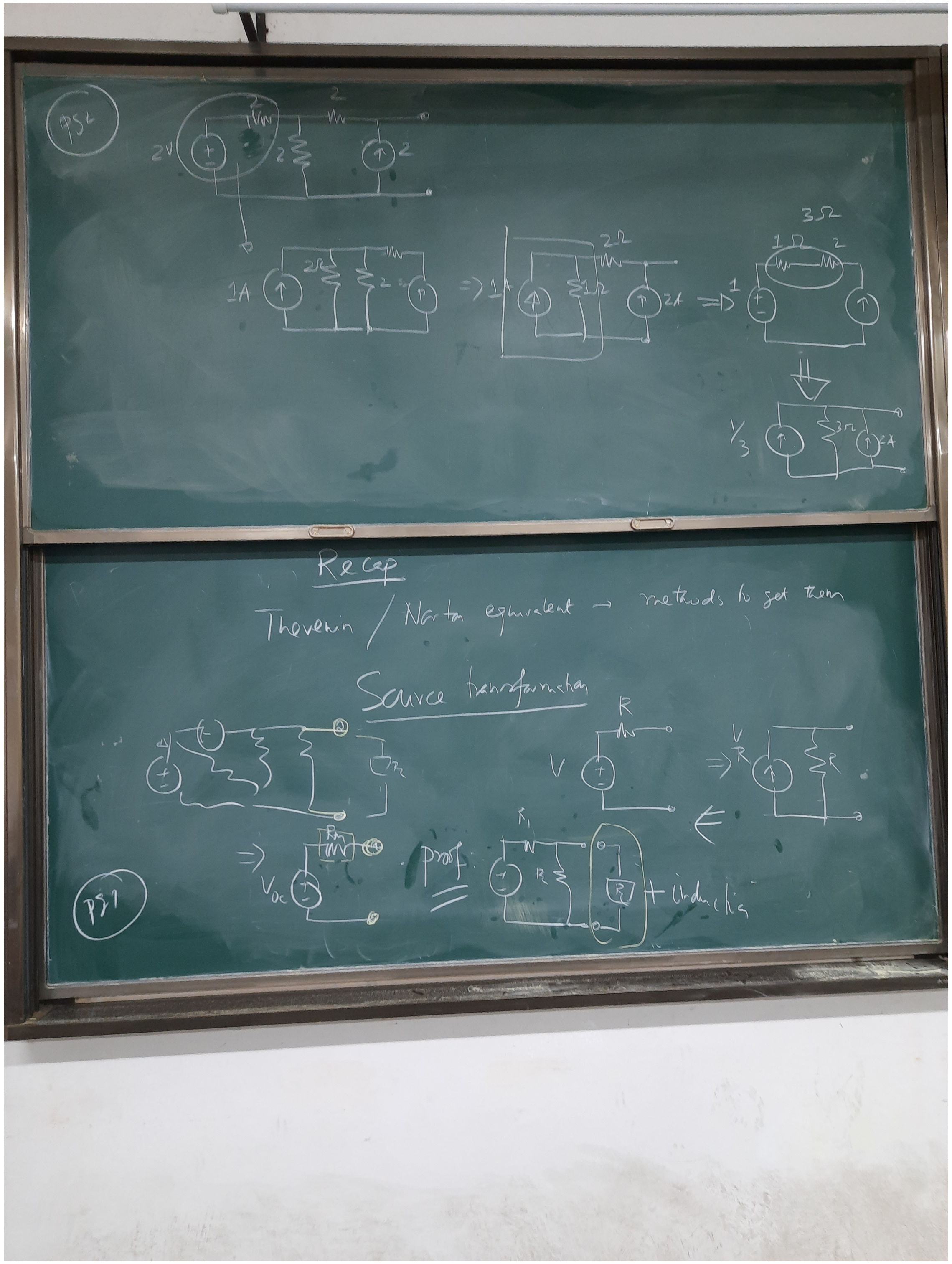
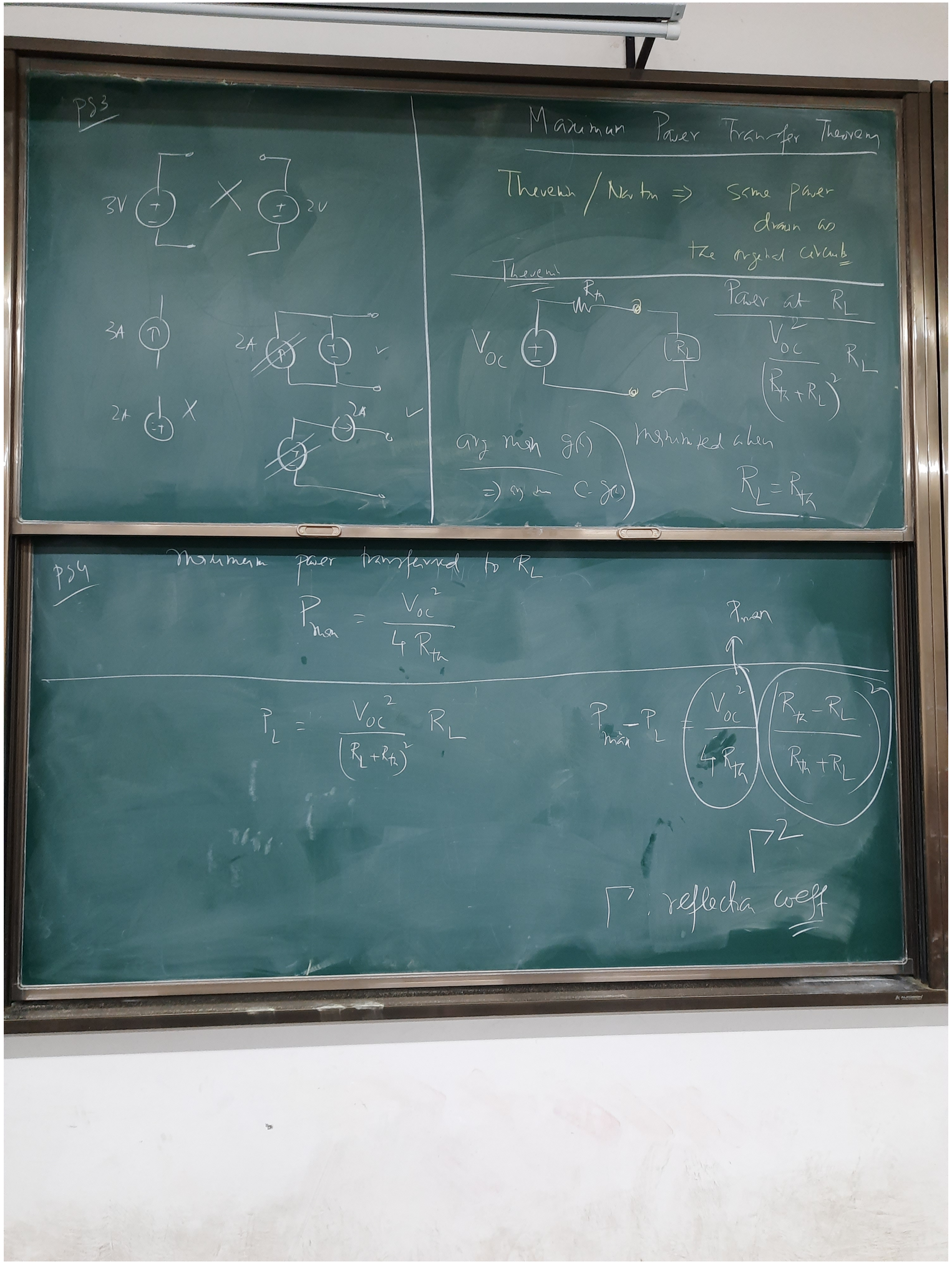


Recap

Thevenin / Norton equivalent → methods to get them

Source transformation



Maximum Power Transfer Theorem

Thevenin / Norton ⇒ same power drawn as the original circuit

Thevenin

Power at R_L

$$\frac{V_{OC}^2}{(R_{th}+R_L)^2} R_L$$

arg max $g(x)$

⇒ arg min $C - g(x)$

maximized when

$$R_L = R_{th}$$

PS4

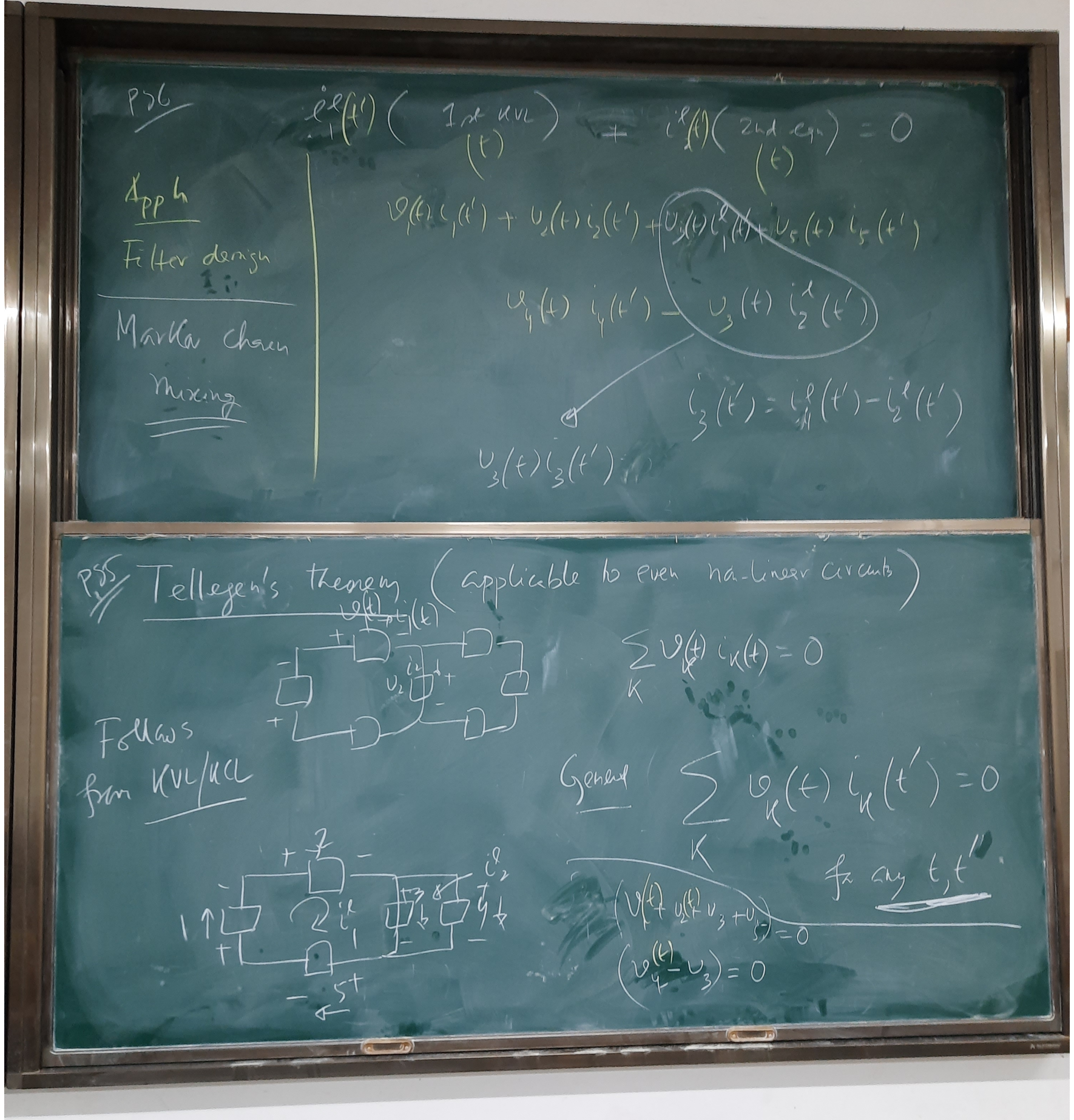
maximum power transferred to R_L

$$P_{max} = \frac{V_{OC}^2}{4R_{th}}$$

$$P_L = \frac{V_{OC}^2}{(R_L+R_{th})^2} R_L$$

$$P_{max} - P_L = \underbrace{\frac{V_{OC}^2}{4R_{th}}}_{P_{max}} \underbrace{\left(\frac{R_{th}-R_L}{R_{th}+R_L}\right)^2}_{\Gamma^2}$$

Γ : reflection coeff



P56

$i_1^{\ell}(t')\left(\begin{matrix}\text{1st KVL}\\(t)\end{matrix}\right) + i_2^{\ell}(t')\left(\begin{matrix}\text{2nd eqn}\\(t)\end{matrix}\right) = 0$

Appln

Filter design

Markov chain

Mixing

$v_1(t) i_1(t') + v_2(t) i_2(t') + v_3(t) i_1^{\ell}(t') + v_5(t) i_5(t')$

$v_4(t)\, i_4(t') - v_3(t)\, i_2^{\ell}(t')$

$i_3(t') = i_1^{\ell}(t') - i_2^{\ell}(t')$

$v_3(t) i_3(t')$

P55 Tellegen's theorem (applicable to even non-linear circuits)

$$\sum_k v_k(t)\, i_k(t) = 0$$

Follows from KVL/KCL

General $$\sum_k v_k(t)\, i_k(t') = 0$$

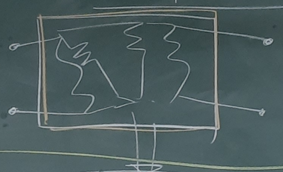
for any t, t'

$(v_1(t) + v_2(t) + v_3 + v_5) = 0$

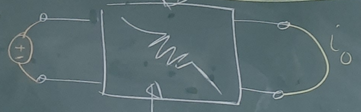
$(v_4(t) - v_3) = 0$

PS7

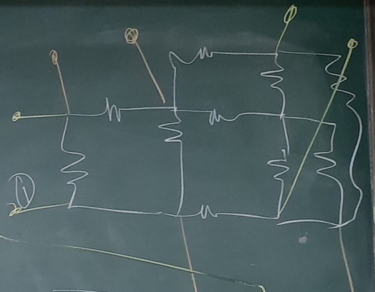
Reciprocity theorems



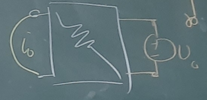
Rec thm part 1



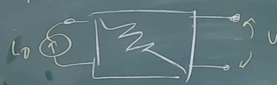
only resistors



then

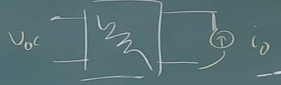


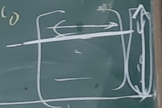
Rec thm part 2



⇒

then





$$\begin{bmatrix} & & \\ & A & \\ & & \end{bmatrix}\begin{bmatrix}\underline{x}\end{bmatrix} = \begin{bmatrix}V\\ \vdots\\ 0\end{bmatrix}$$

$x_1 = A^{-1}\begin{pmatrix}V\\0\end{pmatrix}$

x_n

$$\begin{bmatrix} & A' & \end{bmatrix} = \begin{bmatrix}x_1\\ \vdots\\ x_n\end{bmatrix} = \begin{bmatrix}0\\ \vdots\\ V\end{bmatrix}$$